Computer Organization

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Contents

Chapter 1 Computer Organization

1 Introduction

I Binary and Decimal Prefix

A binary and decimal prefix is a unit prefix that indicates a multiple of a unit of measurement by an integer power of two and ten respectively. They are most often used in information technology as multipliers of bit and byte, in which the short prefixes are prefixed before b (representing bits) or B (representing bytes), and the long prefixes are prefixed before bits or bytes. Originally, the prefixes such as K and kilo are used as binary prefix too. To reduce confusion, IEC now use the prefixes such as Ki and kibi as binary prefixes and reserve the prefixes such as K and kilo as decimal prefixes, while JEDEC still uses them as binary prefixes to units of semiconductor storage capacity.

Binary Value	IEC 60027-2 (short)	IEC 60027-2 (long)	JEDEC JESD100B.01 (short)	JEDEC JESD100
2^{10}	Ki	kibi	K	kilo
2^{20}	Mi	mebi	M	meg
2^{30}	Gi	gibi	G	giga
2^{40}	Ti	tebi	T	tera
2^{50}	Pi	pebi	-	-
2^{60}	Ei	exbi	-	-
2^{70}	Zi	zebi	-	-
2^{80}	Yi	yobi	-	-
2^{90}	Ri	robi	-	-
2^{100}	Qi	quebi	-	-

II Moore's law

Moore's law is the observation that the number of transistors in an integrated circuit (IC) doubles about every two years, with minimal increase in cost. The observation is named after Gordon Moore, the co-founder of Fairchild Semiconductor and Intel and former Chief Executive Officer of the latter.

III Classes of Computers by Applications

i Personal computer (PC)

A computer designed for use by an individual, usually incorporating a graphics display, a keyboard, and a mouse.

ii Server

A computer used for running larger programs for multiple users, often simultaneously, and typically accessed only via a network.

iii Embedded computer

A computer inside another device used for running one predetermined application or collection of software.

iv Personal mobile device (PMD)

A small wireless device that typically connects to the Internet, relies on batteries for power, and in which software is installed by downloading apps. Conventional examples are smart phones and tablets.

IV Programming Language

i Abstraction

Abstraction is the process of hiding unnecessary details while exposing only the essential features needed for a particular task.

ii Instruction set architecture (ISA)

An instruction set architecture (ISA), also called architecture, is an abstract model that defines the programmable interface, the instructions, of the CPU of a computer. Some well-known instruction set architectures are x86-64, ARM, and RISC-V.

An instruction may specify:

- opcode: the operation to be performed, e.g. add, copy, test.
- any explicit operands: e.g. registers, literal values, addressing modes used to access memory.

More complex operations are built up by combining these simple instructions.

Instructions in binary is called machine code or machine language. Instructions in symbolic, human-readable representation is called assembly language, assembly, assembler language, symbolic machine code, ASM, or asm.

An assmebler is a program that translates assembly into machine code.

Low-level languages typically refer to machine languages and assembly languages.

Hardware that obeys an ISA is called an implementation.

iii High-level language

A high-level language is a programming language with strong abstraction from the details of the computer, such as C, C++. Java, or Visual Basic. They are compiled with a compiler or intepreted with an intepreter to machine code.

iv Application binary interface (ABI)

An application binary interface (ABI) is an interface exposed by software that is defined for in-process machine code access. Often, the exposing software is a library, and the consumer is a program.

V Input/output (I/O) device

i Input device

An input device is a mechanism through which the computer is fed information, such as a keyboard.

ii Output device

An output device is a mechanism that conveys the result of a computation to a user, such as a display, or to another computer.

VI Central processor unit (CPU) or processor

i Central processor unit (CPU) or processor

A CPU is the active part of a computer, which contains the datapath and control and which adds numbers, tests numbers, signals I/O devices to activate, and so on.

ii Datapath

The datapath is the component of a processor that stores and transforms data.

iii Control

The control is the component of a processor that decodes the instructions and decides what the datapath should do in each cycle.

VII Operating System (OS)

An operating system (OS) is system software that manages computer hardware and software resources, and provides common services for computer programs.

VIII Memory

i Memory cell

A memory cell is a device, such as an electronic circuit, that stores one bit of binary information and it must be set to store a logic 1 (high voltage level) and reset to store a logic 0 (low voltage level). Its value is maintained/stored until it is changed by the set/reset process. The value in the memory cell can be accessed by reading it.

ii Volatile and non-volatile memory

Data storage, also called computer memory or memory, is called volatile memory if it requires power to maintain the stored information and non-volatile (or nonvolatile) memory (NVM) if it retains data even in the absence of a power source.

iii Main memory

Main memory or primary memory, also called computer memory or memory, is volatile memory in which programs are kept when they are running and that contains the data needed by the running programs. Main memories are usually DRAM now.

iv Random-access memory (RAM)

Random-access memory (RAM) is data storage that can be read and changed in any order.

v Static RAM (SRAM)

Static RAM (SRAM) is volatile RAN that holds its data permanently in the presence of power, typically used as cache memory and internal registers of a CPU. A memory cell of SRAM is a flip-flop that consists of transistors.

vi Dynamic RAM (DRAM)

Dynamic RAM (DRAM) is volatile RAN that decays in seconds and thus must be periodically refreshed, typically used as main memory. A memory cell of DRAM consists of a transistor and a capacitor.

SRAM is faster than DRAM but it is more expensive in terms of silicon area and cost.

vii Secondary memory

Secondary memory or secondary storage is non-volatile memory where systems store data persistently, typically consists of flash memory now and HDDs for older PCs and servers.

viii Flash memory

Flash memory is a type of Electrically Erasable Programmable ROM (EEPROM) optimized to be faster, denser, cheaper, and erasable in blocks or pages. Flash memories usually have built-in programming and erase capability so that data can be written to the flash memory while it is in place in a circuit without the need for a separate programmer.

NOR and NAND flash:

- NOR Flash: Output lines are connected in parallel and cells are erased in blocks and with random access for read. Writing and erasing is slower and cell size is larger.
- NAND Flash: Output lines are connected in series and cells are erased in pages and with sequential access for read only. Writing and erasing is faster and cell size is smaller. Solid-state drive (SSD), USB flash drive, embedded MultiMediaCard (eMMC), Universal Flash Storage (UFS), secure digital (SD) card are common types of storage devices built using NAND flash.

Type of cells:

Type	Bits per cell
Single-level cell (SLC)	1
Multi-level cell (MLC)	2
Triple-level cell (TLC)	3
Quad-level cell (QLC)	4

The less bits per cell, the flash memory is usually faster, more reliable, and more expensive per unit of data storage. TLC and QLC are used in consumer SSDs.

ix Hard disk drive (HDD), hard disk, hard drive, or magnetic disk

Non-volatile secondary memory composed of rotating platters coated with a magnetic recording material.

IX Network

i Computer network or network

A network is a group of communicating computers and peripherals known as hosts, which communicate data to other hosts via communication protocols, as facilitated by networking hardware.

ii Local area network (LAN)

A local area network (LAN) is a computer network that interconnects computers within a limited area such as a residence, campus, or building, and has its network equipment and interconnects locally managed.

iii Wide area network (WAN)

A wide area network (WAN) is a telecommunications network that extends over a large geographic area.

X Performance

i Response time, execution time, wall clock time, or elapsed time

The total time required for the computer to complete a task, including disk accesses, memory accesses, input/output (I/O) activities, operating system overhead, CPU execution time, and so on.

ii Throughput or bandwidth

The number of tasks completed per unit time.

iii CPU execution time or CPU time

The time the CPU spends computing for a task.

iv User CPU time

The CPU time spent in a program itself.

v System CPU time

The CPU time spent in the operating system performing tasks on behalf of a program.

vi Clock cycle, tick, clock tick, clock period, clock, or cycle

The time for one clock period, usually of the processor clock, which runs at a constant rate.

vii Clock rate

The reciprocal of clock period.

viii Clock cycles per instruction (CPI)

Average number of clock cycles per instruction for a program or program fragment.

ix Instruction count

The number of instructions executed by the program.

x Relation

CPU time = Instruction count $\times$ CPI / Clock rate.